

For questions 1-3, identify whether the statement is true or false. If it is false, rewrite the statement to make it true.

1. Line plots are useful to show large quantities of data. *True or False*
2. Histograms and stem-and-leaf plots can show large quantities of data and the overall shape. *True or False*
3. Box plots show the overall shape and information to calculate the 5-number summary. *True or False*
4. A fuel efficiency, in miles per gallon, of 250 cars were surveyed. Which graphical display would best display the data?

- ☐ Box plot
  - ☐ Line plot
  - ☐ Histogram
  - ☐ Stem-and-leaf plot
5. Explain your reasoning for question 4.
6. Greg was in the market for a new SUV. He recorded the fuel efficiency for his top 12 choices.

{22.5, 17.2, 18, 20.2, 13.6, 19.4, 14, 20.4, 11.6, 16, 15.9, 20}

Explain which display would be better for the data, a line plot or a histogram.

A survey was conducted to middle school students in order to determine their favorite type of music. The frequency and relative frequency tables below represent the data collected.

**Frequency Table**

| Type of Music | Frequency |
|---------------|-----------|
| Rap           | 65        |
| Hip hop       | 87        |
| Rock          | 32        |
| Other         | 16        |

**Relative Frequency Table**

| Type of Music | Frequency |
|---------------|-----------|
| Rap           | 32.5%     |
| Hip hop       | 43.5%     |
| Rock          | 16%       |
| Other         | 8%        |

7. In order to create a circle graph, the data from the table is needed.

- ☐ frequency
- ☐ relative frequency

8. In order to create a bar graph, the data from the table is needed.

- ☐ frequency
- ☐ relative frequency

9. Which of the follow are types of displays for univariate numerical data? *Select all that apply.*

- ☐ Frequency Table
- ☐ Box plot
- ☐ Line Plot
- ☐ Histogram
- ☐ Circle Graph
- ☐ Stem-and-leaf plot
- ☐ Bar Graph

10. Which of the follow are types of displays for univariate categorical data? *Select all that apply.*

- ☐ Frequency Table
- ☐ Box plot
- ☐ Line Plot
- ☐ Histogram
- ☐ Circle Graph
- ☐ Stem-and-leaf plot
- ☐ Bar Graph

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Lesson 2 Homework

Name \_\_\_\_\_  
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Identify whether each type of data is categorical or numerical. Then identify if it is univariate or bivariate.

1. The number of houses sold in Broward County in different price ranges.

☐ Categorical  
☐ Numerical

☐ Univariate  
☐ Bivariate

2. The types of houses sold in Broward County.

☐ Categorical  
☐ Numerical

☐ Univariate  
☐ Bivariate

3. The number of houses sold in Broward County different price ranges and the average time the houses are on the market for.

☐ Categorical  
☐ Numerical

☐ Univariate  
☐ Bivariate

Use the following for questions 4-6.

Greg was in the market for a new SUV. He recorded the fuel efficiency for his top 12 choices.

{22.5, 17.2, 18, 20.2, 13.6, 19.4, 14, 20.4, 11.6, 16, 15.9, 20}

4. This data set is ☐ Categorical  
☐ Numerical and ☐ Univariate  
☐ Bivariate

5. Which is the best graphical representation to display the data?

☐ Scatterplot  
☐ Box plot  
☐ Line Plot  
☐ Histogram  
☐ Circle Graph  
☐ Stem-and-leaf plot  
☐ Bar Graph

6. Create a graphical representation for the data chosen in question 5.

The National Weather Center collected data on weather conditions in various locations. Data was collected over a few years comparing the elevation above sea level in feet and the average number of clear days in 12 different cities in the United States.

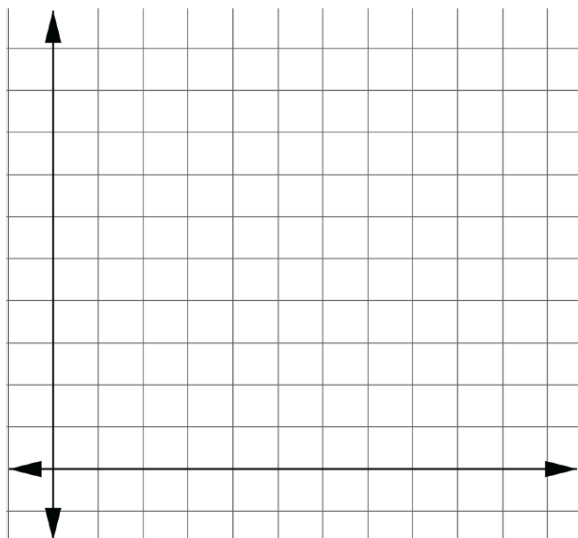
| City               | Elevation above sea level | Average number of clear days |
|--------------------|---------------------------|------------------------------|
| Anchorage, AK      | 102                       | 45                           |
| Boston, MA         | 141                       | 102                          |
| Tampa, FL          | 48                        | 146                          |
| Salt Lake City, UT | 4,226                     | 95                           |
| Charlotte, NC      | 761                       | 127                          |
| Seattle, WA        | 174                       | 78                           |
| Boise, ID          | 2,730                     | 115                          |
| Milwaukee, WI      | 617                       | 90                           |
| New York, NY       | 33                        | 72                           |
| Cincinnati, OH     | 482                       | 111                          |
| Boulder, CO        | 5,430                     | 65                           |
| San Francisco, CA  | 52                        | 198                          |

7. The data shown in the table is
 

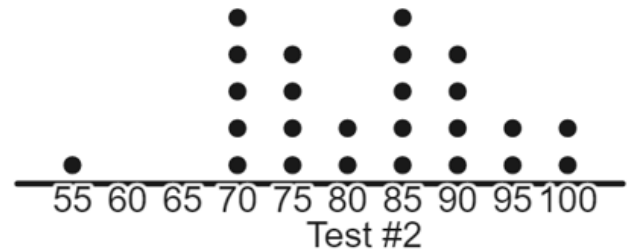
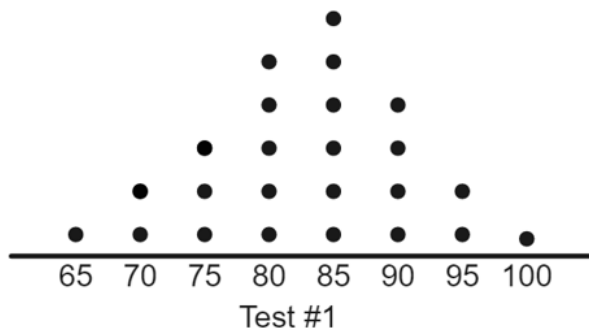
☐ univariate.  
☐ bivariate.
8. The data in the table should be displayed in a
 

☐ line plot  
☐ scatter plot

 .
9. Explain your reasoning from question 8.
10. Create a display to represent the relationship between elevation above sea level and the average number of clear days.

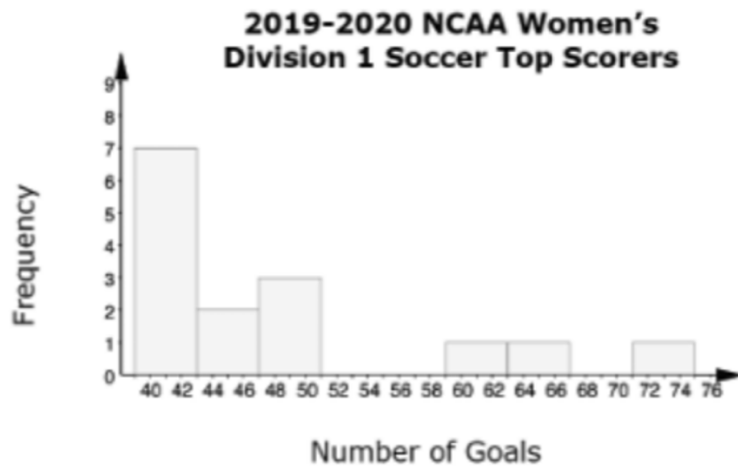


Two dot plots are shown, with results from two recent tests.



1. What is the mean of the scores for Test #1?
2. What is the median for the scores of Test #1?
3. Which measure of center is the most appropriate to use to describe the scores of Test #1? Explain how you know.
4. Find the interquartile range of Test #1 and interpret it in the context of the problem.
5. What is the mean of the scores for Test #2?
6. What is the median of the scores for Test #2?

7. Which measure of center is most appropriate to use to describe the scores of Test #2?
8. Write a sentence comparing the results of Test #1 to Test #2 using measures of center and measures of spread.
9. In the histogram shown, which measure of center would best be used to describe the data? How do you know?



10. A table of values for the heights of the students in a class has a mean of 61 inches and a standard deviation of 1.4 inches. Interpret the variation in height.

Algebra 1

Unit 12

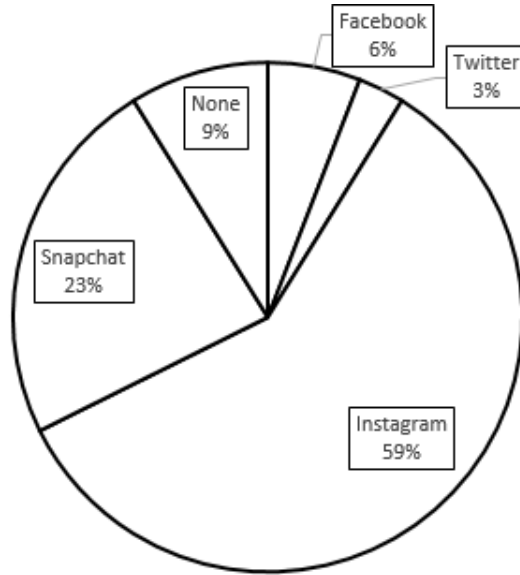
Lesson 4 Homework

Name \_\_\_\_\_

Date \_\_\_\_\_

Period \_\_\_\_\_

A circle graph is shown, with the results of a survey of a classroom of 34 9<sup>th</sup> grade students about which social media platforms they use.



1. What is the most frequently used social media platform?
2. What is the least frequently used social media platform?
3. How many students in the class used Snapchat?
4. How many students in the class do not use any social media?
5. How many more students use Instagram than Facebook?

A table is shown with the results of a survey of a group of 110 students about their favorite type of movie.

| Favorite Movie Genre | Number of Students |
|----------------------|--------------------|
| Comedy               | 42                 |
| Action               | 33                 |
| Romance              | 12                 |
| Mystery              | 23                 |

6. Create a bar graph showing the results of this survey.
7. Create a relative frequency table showing the results of this survey.
8. Create a circle graph showing the results of this survey.
9. What percentage of students preferred either mystery or action movies?
10. What percentage of students did not prefer comedies?



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Lesson 5 Homework

Name\_\_\_\_\_

Date\_\_\_\_\_

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A study shows that the number of times a student skips breakfast has a negative correlation with student test scores.

1. What does it mean that there is a negative correlation?
2. Does this mean that not eating breakfast causes students to do poorly on tests?
3. What other factors might contribute to this association?
4. What other information would be helpful to determine whether not eating breakfast causes lower test scores?

At a theme park, there is a positive correlation between the amount of rainfall and the number of ponchos sold.

5. What does it mean that there is a positive correlation?
6. Does this mean that increased poncho sales leads to more rainfall? If not, what does it mean?
7. What other factors might contribute to this association?

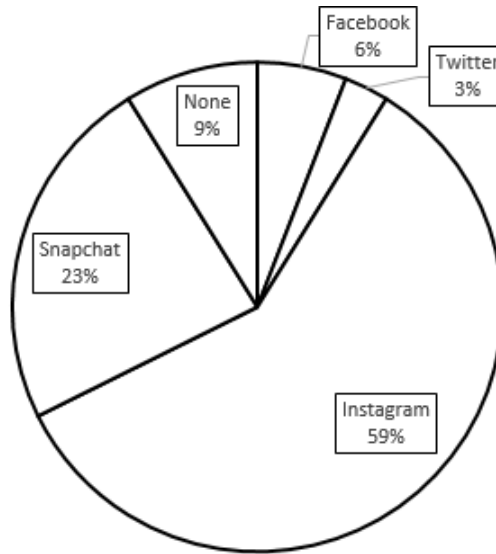
At a ski resort, as snow accumulation increases, so does the injury rate.

8. What kind of correlation does this association represent?
9. Is there evidence of a causal relationship?
10. What other information would be helpful to evaluate the claim of a causal relationship?

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Lesson 6 Homework

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

A circle graph is shown, with the results of a survey of 9<sup>th</sup> grade students about which social media platforms they use, with a margin of error of 1.2%. There are 475 9<sup>th</sup> graders in a particular school.



1. Based on this survey, what is the estimated number of 9<sup>th</sup> graders who have an Instagram account?
2. Based on this survey, what is the minimum number of 9<sup>th</sup> graders that would be expected to have a Facebook account?
3. Based on this survey, what is the maximum number of 9<sup>th</sup> graders that would be expected to have a Twitter account?
4. Based on this survey, what is the range of 9<sup>th</sup> graders that would be expected to have a Snapchat account?
5. Based on this survey, what is the range of 9<sup>th</sup> graders that would be expected to have no social media accounts?

A survey of parents indicated that 62% of parents felt their students spent too much time on social media, with a margin of error of 2%.

6. Out of a group of 2400 parents, how many parents would likely feel that their children spend too much time on social media?
  
  
  
  
  
  
  
7. Out of a group of 2400 parents, how many parents would likely **not** feel that their children spend too much time on social media?

In a survey of 200 families, 32% of families reported that they eat out less than twice a week, 45% report they eat out twice a week, and 23% report they eat out more than twice a week. The survey has a margin of error of 0.8%.

8. What is the expected number of families that eat out twice a week?
  
  
  
  
  
  
  
9. What is the range of families that eat out less than twice a week?
  
  
  
  
  
  
  
10. What is the range of families that eat out more than twice a week?